GEK-86255A

DRIVE SYSTEM INSTRUCTIONS

STATIC KRAEMER DRIVE SYSTEM With DIRECTO-MATIC® PLUS HARDWARE

GE REQUISITION 541-27816

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

STATIC KRAEMER
DRIVE SYSTEM

(GEK-86255A)

ISSUED: February 1982
REVISED: August 1982

Plains Electric G. & T. Co-op.

Unit 1
Sulfur Dioxide Absorption Equipment

Burns & McDonnell

File A134(G)
Project No. 78-140-1
Contract No. 144

Combustion Engineering, Inc.

Contract No. 05780
Purchase Order No. 2000037
Requisition No. 52401

General Electric Company

Requisition No 541-27816
M.L. No. 401X105 - Main SKD Drives
M.L. No. 306X181 - Standby Contactors
M.L. No. 306X182 - Standby Resistors

The information herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, Drive Systems Department, USA.

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DEVICE NOMENCLATURE	ELEMENTARY	SHEET NO.
Pulse Amplifier Trans	DS3800HPTF	Shs. 2,4
Neon Light and Attenuator	DS3800NTVB	Shs. 2,4
High Level Isolated Input	DS3800HSCD	Shs. 2,4
Logic Element	IC3800HLEA	Shs. 2,4
Gain Programming	DS3800NGPA	Shs. 2,4
Signal Isolator	DS3800NISA	Shs. 2,4
Operational Amplifier	DS3800NOAA	Shs. 2,4
Speed Current Regulator	DS3800NSCA	Shs. 2,4
Firing Circuit	DS3800NFCC	Shs. 2,4
Firing Circuit Option 'B'	DS3800NFOB	Shs. 2,4
Firing Circuit Option 'C'	DS3800NFOC	Shs. 2,4
Firing Circuit Option 'D'	DS3800NFOD	Shs. 2,4
Bridge Fanout	DS3800XBFA	Shs. 2,4
Thyristor Fanout	DS3800XTFA	Shs. 2,4
Plug-in Relays (KMRB)	IC3600KMRB	Shs. 3.0-3FL.
Tach. Pulse Rate Converter	DS3800NTRA	Shs. 2,4
Current Balance Regulator	DS3800NCCA	Shs. 2,4
Power Supply	DS3820PS5A	Shs. 2,4
P105V Power Supply	259A7739G1	Shs. 1,3
Fast Drop-Out Module	DS3820FDRB	Shs. 2,4
KMR Relays	IC3603A177	Sh. 3

8 DEVICE PUBLICATIONS

DEVICE	DRAWING
Drive Test Module (DTM)	GEK-65150A
Drive & Direct Control Equipment	GEK-61680
Guidance Service Instructions	GEK-61700
SB1 Switch	GEH-908
CR2940 Lights	GEH-1052, GEJ-2836
Noise Control	277A6956, Sheet 1,2
Grounding	277A6928, Sh. 1

RELAYS & CONTACTORS:

CR209B022ECA	GEH-3619
IC2800 1178 - 1180	GEH-751
IC2810A14	GEH-2385
IC2820A100, C100	GEH-1941
Motor Control Center	GEH-2614
Starters, Type 206D	GEH-3606
Grounding	277A6928, Sh. 1

9 AIR HANDLING SYSTEM INSTRUCTIONS

See Index Under TAB 9

10 AIRPAX TACHOMETER PICKUP

Outline	087-304-0001, Sheet 1
Specification	087-304-0001, Sheet 2

11 Isolation Transformer Publication

Federal Pacific Switch	IN-825.1

Air-Core DC Smoothing Reactor Publication

INTRODUCTION

These instructions apply to General Electric-supplied AC adjustable-speed AC motor drive systems for the No. 1 Generating unit at Plains Electric Escalante Generating Station, Pruitt, New Mexico.

A list of equipment furnished is provided under TAB 3 of this book. All the equipment furnished will be within the control house (Item 15) except for the motors (Item 1) and the Standby Contactor/Resistor assemblies (Item 5).

PROPRIETARY INFORMATION OF GENERAL ELECTRIC COMPANY

GENERAL INSTALLATION INSTRUCTIONS FOR AC SYSTEMS

Introduction

When the equipment is installed, the interconnection and elementary diagrams should be closely followed. The recommended level wiring practices for interconnecting wire and cable must be followed to reduce the possibility of electrical noise.

The control panels, cabinets, desks, rotating equipment, transformers, reactors, and individual components have been tested at the factory, but they have not been operated as a complete system. The resistors in the various circuits have been set at values based on calculations from data available, or from previous tests of similar equipment. It will be necessary, therefore, to make further adjustments when the equipment is placed in operation. Before proceeding with the various adjustments, care should be exercised to see that each resistor and rheostat is set at the approximate value indicated on the elementary diagram.

Mechanical Operation

After the rotating machines have been installed, they should be checked for mechanical alignment and clearances by turning each machine over manually or with the shop crane. Check the drive for proper direction of rotation. For details regarding the installation of machines, refer to the applicable standard device instruction.

Meters and Instruments

For making the necessary checks and adjustments, suitable equipment must be available to accomplish the work in an accurate and satisfactory manner. The following equipment should be available for adjustment of the control system:

1. A portable volt-ohm-milliammeter.
2. A portable DC voltmeter.
3. A tachometer or RPM counter and stop-watch.
4. A portable 6-channel recorder such as the Brush oscillograph, with amplifier.

   NOTE: Such a recorder is extremely useful in tuning up equipment involving complex regulating systems.

5. Dual-trace oscilloscope, such as the Tektronix Model 516.

PROPRIETARY INFORMATION OF GENERAL ELECTRIC COMPANY

Original voltage and polarity checks should be made with the portable instruments connected directly to the points under investigation rather than the panel-mounted instruments. By this method, any errors in instrument connections can be found and corrected, avoiding erroneous indications caused by incorrect instrument connections. After it has been ascertained that a panel meter is indicating the correct polarity, it should be so marked, and may be used thereafter.

PROPRIETARY INFORMATION OF GENERAL ELECTRIC COMPANY

SYSTEM PARAMETERS

Speed Range

Operating Speed Range 50 - 95.5% of synchronous speed in Kraemer Mode
Starting Speed Range 0 - 50% (By AC Resistors)
Standby Speeds of 50%, 69% and 79% of synchronous speed

Acceleration

1% of required speed per second (Adjustable)
Maximum acceleration rate will be limited by 115% maximum torque availability.

Altitude and Temperature

Equipment is based on operation at a maximum altitude of 6900 feet with a maximum temperature of 40°C and a minimum temperature of 0°C.

Outdoor equipment suitable for operation down to -35°C.

Bus Voltages

4160V AC	60 Hz	3 Phase
480V AC	60 Hz	3 Phase
125V DC	Battery Voltage	

PROPRIETARY INFORMATION OF GENERAL ELECTRIC COMPANY

SYSTEM DESCRIPTION

The AC Adjustable-speed AC Motor Drive systems described herein are for two induced draft fans on the No. 1 Generating Unit at Plains Electric Escalante Generating Station, Prewitt, New Mexico.

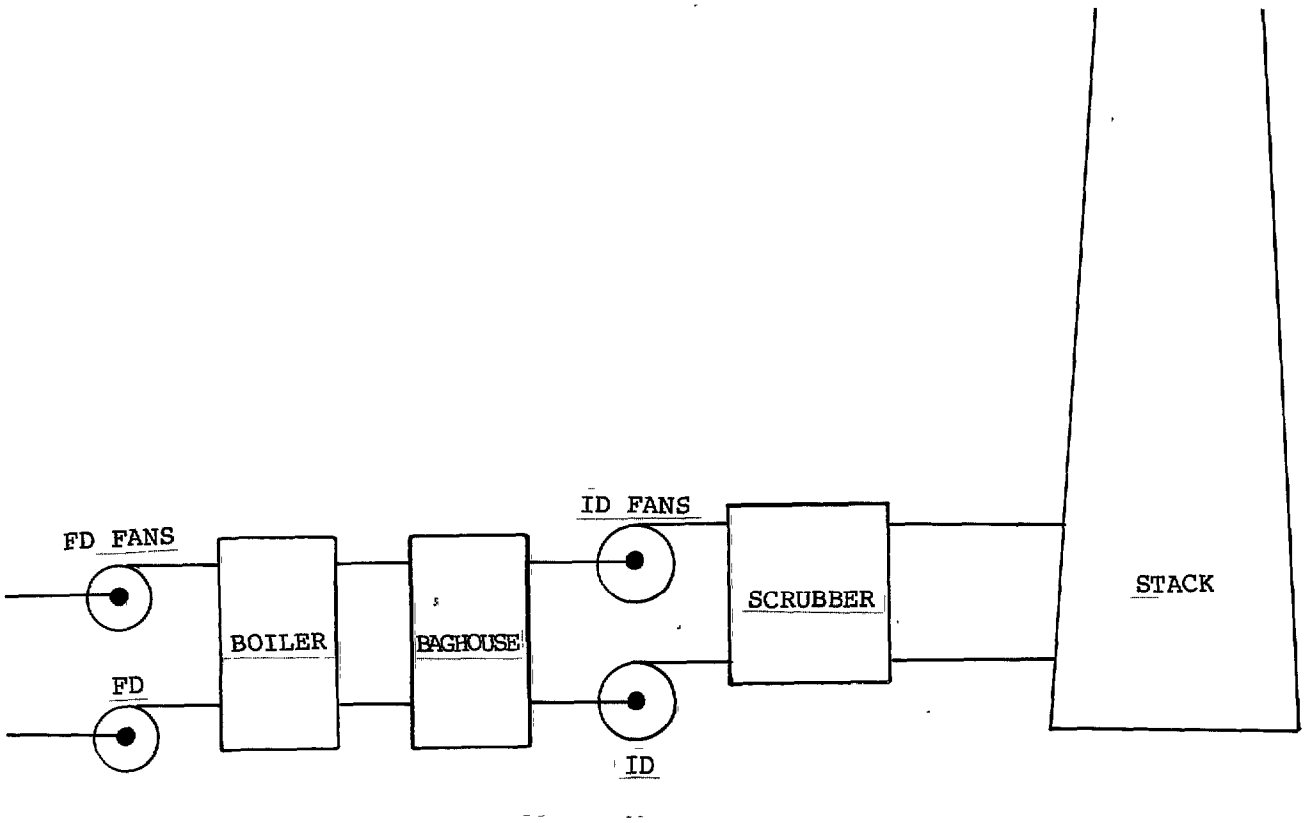


Figure 1
Simplified Boiler - ID Fan Arrangement

PROPRIETARY INFORMATION OF GENERAL ELECTRIC COMPANY

The power circuit is shown on sheet 0L of the drive elementary diagram.

The auxiliary system for the generating unit is shown on sheet 0M of the elementary diagram.

The Static Kraemer Drive is a package of controls and subassemblies arranged to form a power recovery system for a wound rotor motor. It is a solid state secondary controller. Other equipments which are interfaced with the Static Kraemer Drive are (1) the Primary Controller - LIMITAMP® or Circuit-Breaker, (2) Auxiliary starting equipment such as Liquid Rheostat or Steel Resistors.

The typical thyristor (SCR) power conversion equipment supplied on Static Kraemer Drives consists of one or more line-ups which are shipped separately from the Salem, Virginia factory. These line-ups, also referred to as control panels, are interconnected with each other and with the motor at the customer's plant. They may be abutted against a switchgear line-up.

The typical equipment is composed of several discrete modules or groupings of control equipment. An example is the firing circuit module which is located in a swing-out "Page" and the Power Bridge which is located in a cabinet.

THEORY OF OPERATION

The Drive System is an adjustable speed drive which recovers motor rotor slip energy and returns it to the drive. This provides a sustained high effeciency over the important part of the operating range.

The Drive System consists of a wound rotor induction motor, a switched resistor, a diode rectifier section, a reactor, a thyristor inverter section, an isolation transformer, and DIRECTO-MATIC® speed regulator and logic control. Both the diode rectifier section and thyristor inverter sections consist of three-phase bridges which have become the standard in large industrial DC power converters. The use of switched resistors permits optimum economic selection of the conversion equipment based on speed range requirements. The reactor is used to obtain smooth current flow.

The speed control of the drive is accomplished by controlling the firing angle of the thyristor inverter. The tachometer measures drive speed. Accurate speed regulation is achieved with a DIRECTO-MATIC® speed regulator that automatically controls the firing of the inverter to maintain set speed. The operating speed of the drive is set or adjusted with a speed reference signal locally or from a process signal.

The power available at the slip rings of the wound rotor motor is first converted to direct current, inverted into alternating current at power system frequency, transformed to match the motor primary voltage and returned to the drive. Most of the motor slip power loss is converted and applied to the motor primary, thus improving drive efficiency.

PROPRIETARY INFORMATION OF GENERAL ELECTRIC COMPANY

Since torque is essentially proportional to motor current, the AC wound rotor motor needs only the current necessary to meet the process acceleration requirements. Instead of the 600 to 700 percent current inrush typical of squirrel cage motors, the Drive System starting current can be as low as 100 percent. The net result is: (1) voltage drop is minimal during start resulting in no disturbance to other electric equipment on the same power system, (2) permits optimized selection of distribution equipment, and (3) results in low mechanical impact on rotating equipment.

Sequencing and Logic

The lighted push buttons and indicating lights which will be mounted on the customer's boiler control panel are shown in Sketch 1, Sheet 1-5.

These are master specialties, Series 10E lighted push buttons and indicating lights having solid or split fields.

Lights will be 28 volts AC and will be powered by customer's transformer.

The control will be designed to allow local running of the drive by a technician during startup and checkout of the equipment. This will involve connecting a few jumpers and using the small diagnostic module furnished with the drive.

The control will provide standby operation of the drive at 50 percent, 69.6 percent, and 79 percent speed. Control logic shall be designed to automatically select the nearest standby run speed point in the event of a fault in the KRAMERSTAT system. Some KRAMERSTAT faults will cause main breaker to trip. Indication of such faults will be provided in the electronics module.

The control will interrogate the customer's input contacts with 120V AC.

The Lockout/Remote Start selector switch on the equipment door can be used to disable remote starting of the drive in the standby mode for both "manual" and "fault transfer from the KRAMERSTAT operation". An interlock off this switch will be in series with the standby ready light.

Standby resistor blowers are normally started and stopped by the boiler operator. However, if the drive transfers from KRAMERSTAT to standby due to a fault, then the blowers will start up automatically. A fan running contact based on tachometer speed in the KRAMERSTAT mode and based on switchgear and contactor position in the standby mode (N.O.) shall be provided for the customer's damper control. A second contact for indication of the same functions shall also be provided for customer's use. The customer also needs a "drive in KRAMERSTAT mode" contact to allow automatic speed control by the combustion control system.

PROPRIETARY INFORMATION OF GENERAL ELECTRIC COMPANY

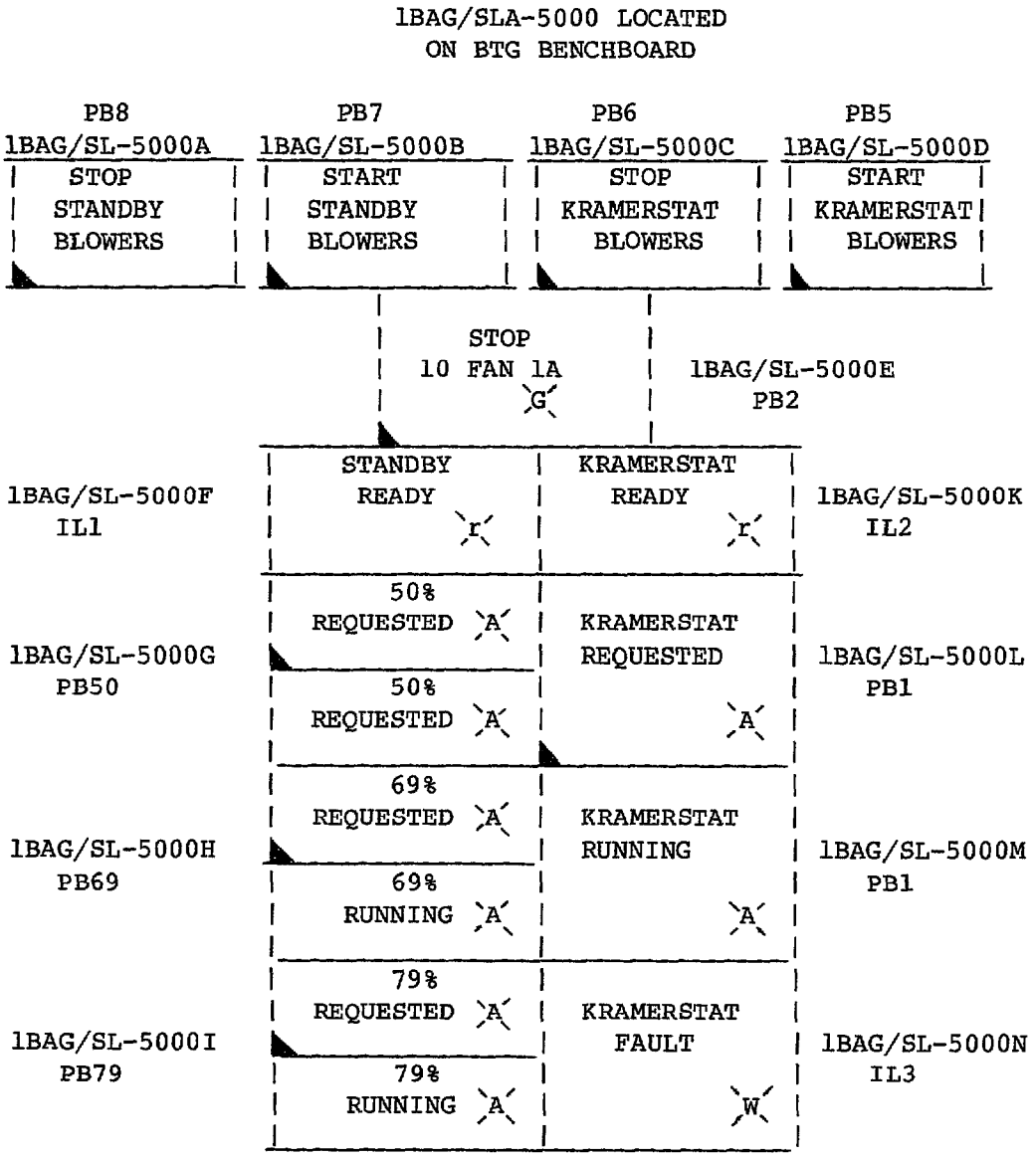
The drive is capable of riding through, with loss of torque, for no more than three seconds on auxiliary bus transfer consisting of:

a. Complete loss of voltage for six cycles (fast transfer)

b. Complete loss of voltage for about one second (slow transfer or residual voltage transfer).

A detailed description of bus transfer operation is contained in paragraphs 4-10 and 4-11 of GEK-83782 under Tab 6.

A step by step description of operator interface with the drives is contained on sheet OG and OH of elementary diagram 228B3432.



Sketch 1

PROPRIETARY INFORMATION OF GENERAL ELECTRIC COMPANY

Nomenclature	Legend	Function
PB2	Stop	Stops drive in all modes.
I11	Standby Ready	Indicates that drive can be started if it is at rest or that it can transfer from the KRAMERSTAT mode to the standby mode if it is runnng in KRAMERSTAT.
I12	KRAMERSTAT Ready	This light will come on when the drive can be transferred (using PB1) to the KRAMERSTAT mode from any one of the three standby speeds.
I13	KRAMERSTAT Fault	This light will come on if a fault exists in the KRAMERSTAT which would prevent the drive from operating in the KRAMERSTAT mode.
PB50 PB69 PB79	Request 50% Speed Request 69% Speed Request 79% Speed	These lighted push buttons are used to request indicated speed in the standby mode. They will be active when the drive is running in any other standby speed, in, KRAMERSTAT, or stopped.
PB50 PB69 PB70	50% Speed Reached 69% Speed Reached 79% Speed Reached	Each light will be lit when the drive has reached indicated speed if that speed has been requested. (Light will not come on when accelerating or decelertaint through a speed.)
PB1	KRAMERSTAT Request	This push button is active only when drive is running at one of the standby speeds and KRAMERSTAT ready light is lit.
PB1	KRAMERSTAT Running	Indicates that drive is running in the KRAMERSTAT mode.
PB5 PB6	Start Stop	Give operator capability of starting stopping blowers in KRAMERSTAT.
PB7 PB8	Start Stop	Give operator capability of starting and stopping blowers in Standby Resistors.

PROPRIETARY INFORMATION OF GENERAL ELECTRIC COMPANY

Signals Required from Drive to Boiler Control

Standby Ready

KRAMERSTAT Ready
50%, 69%, 79% Speed Reached (3 Signals)

KRAMERSTAT Running (Two)

KRAMERSTAT Fault

Motor Speed (0-10V DC)

Rotor Current

Interlock (N.O.) Permissive to Open Damper (Two)

KRAMERSTAT Blowers Running

Standby Blowers Running

(See Also Alarm Indications)

Signals from Boiler Control to Drive

50%, 69%, 79% Standby Speed Request (3 Signals)

KRAMERSTAT Request

Speed Set (4-20 ma)

Stop

KRAMERSTAT Blowers, Start/Stop

Standby Resistor Blowers, Start/Stop

Signals Between Drive and Switchgear

Close Breaker

Trip Breaker

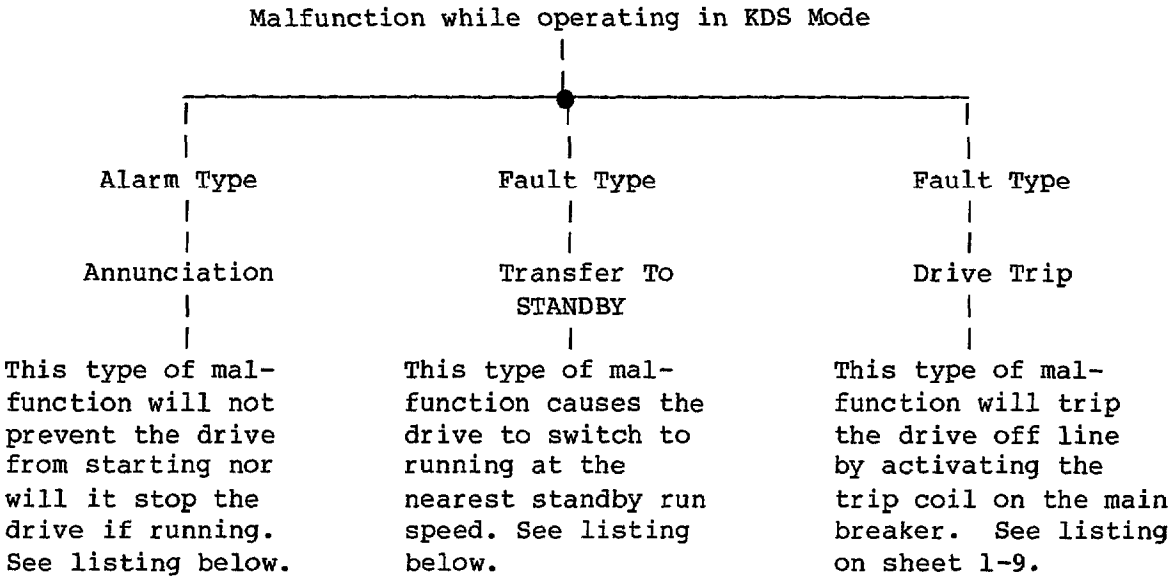
Auxiliary Interlock to Indicate Breaker Closed

(See also Thermostats in Control Equipment House - Item 15)

Signal From Unit Protective Relays To Drive
Auxiliary bus open - Automatic transition from unit auxiliary to start-up transformer.

PROPRIETARY INFORMATION OF GENERAL ELECTRIC COMPANY

Drive Protection and Annunciation



Alarm Indications

The following alarm type malfunctions send an alarm signal to customer's annunciator:

a. Building overtemperature (Low setting).

b. Clogged ventilation filters.

c. Loss of DC power to building thermostats.

Malfunctions which will cause a transfer to standby

The following fault type malfunctions will cause the drive to transfer to the standby mode. A contact is also provided for customer's annunciator.

a. Loss of 480V power supply
b. Loss of power electronics cooling equipment
c. Reactor overtemperature
d. Transformer overtemperature
e. Failure of multiple series diodes or suppression of the inverter
f. Failure of electronics power supply
g. KRAMERSTAT FAULT
h. Building overtemperature (high setting)

PROPRIETARY INFORMATION OF GENERAL ELECTRIC COMPANY

Malfunctions which will cause main breaker to be tripped

a. Loss of standby cooling if running on standby resistors.

b. Incomplete sequence of assignment contactors

c. AC greater than DC for more than one cycle

d. Stall during starting for more than 2 seconds

PROPRIETARY INFORMATION OF GENERAL ELECTRIC COMPANY

WOUND ROTOR MOTORS

Wound rotor motors have been applied to adjustable speed ac drives for more than 50 years.

The stator of such a motor is very similar to that of a squirrel cage motor or a synchronous motor.

There are certain factors which should be kept in mind regarding the use of wound rotor motors on Static Kraemer drives.

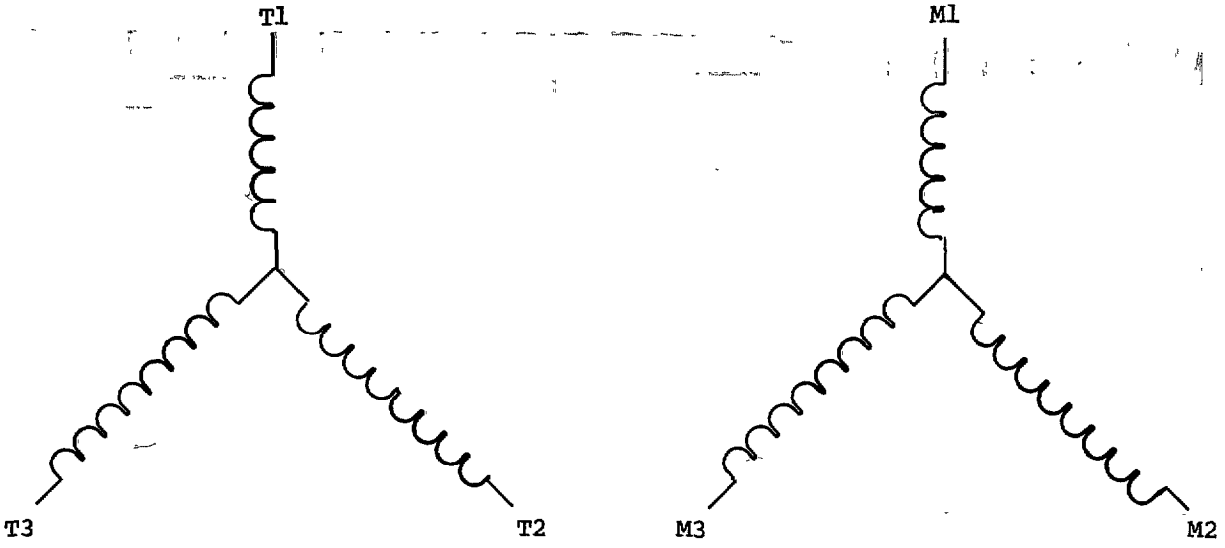
1) The frequency and voltage in the rotor is proportional to slip.

2) Rotor frequency and voltage are maximum at standstill.

3) If the motor is self cooled then there is a minimum speed below which it cannot carry full torque load. In general, the motor can handle a load having torque declining proportional to speed squared without difficulty.

4) In order to provide adequate bearing lubrication the motor cannot be operated below about 30% speed for long periods.

5) Motors are often shipped without oil in the bearings. Be sure to fill bearings with the oil specified by the manufacturer before starting the motor. Light turbine oil is typical.



STATOR ROTOR

Rotation of motor when looking at the end opposite drive end is normally CW for phase rotation 1, 2, 3 applied to stator terminals T_1, T_2, T_3.

PROPRIETARY INFORMATION OF GENERAL ELECTRIC COMPANY

LIST OF EQUIPMENT

Item	
1	Motor
2	Power/Control Panel
5	Standby Run Contactors and Resistor Cabinet
3	Reactor
4,7	Dry-Type Transformer
6	Capacitors
8	Drive Test Module
15	Control Equipment House
16	Disconnect and Switching Devices
17	Iron Core Filter Reactors
18	Single Phase Static Inverters

Item 1 Qty 2 — Motor

Wound rotor motor, type M. rated 4500 HP, 885 RPM, 4000 volts, 3 phase, 60 Hertz, Class B insulation, suitable for non-reversing service, and will have the following ratings with rated voltage and frequency at the stator terminals.

Secondary voltage 1260

Ratings

Temperature Rise	STD
Service Factor	1.0
At ambient temperature	40°C to -35°C
And altitude	6900 feet
Maximum Torque	175 percent breakdown

Construction

Mounting	Horizontal
Shaft	Single Extension
Bearings (non-thrust)	Sleeve
Enclosure	Weather Protected Type II
Frame Size	6380
Hardware	Corrosion Resistant

Features

Bearing Thermocouples
Space Heaters
Stator RTDs, 10 ohm.
Oversize Conduit Boxes

Accessories

Current Transformers for measurement of stator current and for differential protection.
Surge Capacitors
Vibration Pads for Vibration Monitoring Equipment
Toothed Wheel and Tachometer Pickup

PROPRIETARY INFORMATION OF GENERAL ELECTRIC COMPANY

Item	Qty.	
2	2	Power/Control Panel Equipment shall be designed for indoor service consisting of the following: 1 - Diode rectifier to rectify wound rotor motor secondary voltage. 1 - Dual converter, rating and protection based on the following: Supply Voltages Power: 4000 volts, 3 phase, 60 hertz Control: 480 volts, 3 phase, 60 hertz 125 Volts DC Rating Power converter selected for 4500 HP, 4000 V motor at speed range of 50 percent. Projected fan HP is 4154 HP at test block speed of 860 RPM. MCR is 2060 HP at 695 RPM. Rating is such that MCR could be accomodated by half the total complement of thyristors. Duty Class 100 percent RMS load continuous from 0 to rated volts 150 percent load for 30 seconds following no-load, such that 1.0 RMS is not exceeded. 40°C maximum ambient 1 - Lot of control equipment including - Current limit - adjustable - Speed regulator (1 percent or better) - Acceleration rate regulator (adjustable) - Speed set by separately mounted operator potentiometer in manual mode of operation and if required by isolated remote 0-10V DC or 4 to 20 ma signal from process control in automatic mode of operation. - High accuracy current regulator - Fault Indication - Ventilation as required 1 - Lot of assignment contactors to provide for assigning motor secondary to either the KRAMERSTAT or the Standby Run equipment. A manual isolating means will also be provided between the diode bridge and the assignment contactor.

PROPRIETARY INFORMATION OF GENERAL ELECTRIC COMPANY

Item Qty.

Additional features:

1. Series redundancy in the power semiconductors (both rectifier and inverter unit).

2. Fuseless power circuit design.

3. Air-cooled.

4. Bus transfer ride-through with power conditioner as described in the AC power section of this specification.

5. Standby-run on resistors with transfer to nearest speed point upon speed control system fault.

6. Manual isolation for servicing the inverter while the drive is in the standby mode.

7. Devices included and mounted on equipment door:

   Drive running on inverter mode light
   Drive running on standby mode light
   Fault light
   Voltmeter - DC loop volts
   2 Ammeters - DC loop amps for each converter
   Speed indicator
   "Lockout Standby - Normal - Lockout SKD" 3-position selector switch. Lockable.

8. Space heaters 115V, 60 Hz are to be provided in the assignment contactor cabinet and the DC contactor cabinet only.

5 2 Standby Run Contactor and Resistor Cabinet

Outdoor, blower ventilated contactor and resistor cabinet to provide three continuous speed points as follows:

% of Synchronous Speed	50	69.6	79
Actual Speed RPM	450	627	711
Horsepower	594	1477	2143

Note that no speed regulation is provided with these speed points. Instead, resistance in rotor secondary circuit is selected so that if the fan horsepowers required at each of the given speeds are as shown, the resistors will allow secondary current to flow to provide torque corresponding to those horsepowers. In other words, if fan torque required is different from those indicated, fan speed will be different.

PROPRIETARY INFORMATION OF GENERAL ELECTRIC COMPANY

Item Qty.

These same resistors will be used to accelerate the drive to the selected speed point.

Blowers can be turned on and off by the boiler operator.

Space heaters are to be provided in the contactor cabinet only - 240V, 60 Hz, single phase.

3 2 DC Link Reactor

Sets of dual reactors for connection between diode rectifier and thyristor power converters. Each smoothing reactor shall be dry-type, suitable for indoor use, and provided with connections to the control system. Reactor to be equipped with an overtemperature detector in each coil wired to a terminal board. Reactor will operate at an altitude of 6900 feet. Reactors to have bifilar windings. Reactors to be in aluminum enclosures.

4 2 Transformer, Dry-Type (For Remote Mounting)

Indoor, self-ventilated, dry-type, rated 150°C continuous for ANSI (T1) duty class, rated 2476 KVA, primary 4000 volts, secondary rated 760 volts, 60 hertz.

Duty cycle 100%

Rating based on maximum ambient (40)°C, 6900 feet altitude.

XC = 10%

Standard Accessories:

Sheet metal enclosure with removable side panels, side ventilation openings.

Bolted panel for access to primary tap terminal board.

Winding temperature detection equipment with alarm contacts.

Provision for jacking and lifting.

Base designed for skidding or rolling in either direction.

Ground pad at low-voltage end.

Pads for isolation of core and coil vibration.

Delta primary delta and wye secondary (12-pulse converter) (3 Winding)

PROPRIETARY INFORMATION OF GENERAL ELECTRIC COMPANY

Item Qty.

High-voltage connection: To isolation switch

Two 2 1/2 percent rated KVA primary taps above and below rated voltage located on terminal board with bolted access panel.

Low-voltage connection: Internal

Control connections: Terminal board on top of core.
Paint color: Indoor ANSI 61

Customer's primary voltage variation will be from 3800 to 4400 volts.

7 A 3-pole, nonload-break isolation switch shall be provided for each transformer from the 4160 volt feeder. The isolation switch shall be mounted on the transformer and shall be provided with cable connections to the transformer primary and shall be provided with terminals for the owner's cable connections. A space heater is to be included (240V) in the switch end drive. Each transformer to have three current transformers in the primary leads to provide overload protection with leads brought out to terminal boards. These terminal boards shall be short circuiting type General Electric Type EB-27 or equal, with terminals identified by current transformer and phase. The isolation switch is to have 3 current limiting fuses connected on one side to the input terminals and on the other side to terminals outgoing to the power factor capacitors described in Item 6.

6 2 Power Factor Correction Capacitors

900 KVAR, 4000 volt capacitors are included to correct the power factor on each system to .73 at 77 percent speed. They are connected to the primary side of the inverter transformer. Capacitors to be equipped with discharge resistors to reduce capacitor voltage to less than 50 volts within 5 minutes after the capacitors are disconnected from the power supply.

Capacitors to be equipped with current limiting fuses for protection of the individual cans.

PROPRIETARY INFORMATION OF GENERAL ELECTRIC COMPANY

Item	Qty.	
15	1	Control Equipment House

Control equipment house having the following features erected on customer's concrete pad or foundations:

Insulated
Forced ventilated
Heated (Space heaters will be provided to bring the temperature to 0°C when none of the electrical equipment in the room is operating.) Once equipment is operating, it will normally dissipate sufficient heat to keep buiding above 0°C.

All equipment except the two motors and the standby resistors and accelerating contactors will be housed in this control house.

Building Overtemperature Thermostats

Two overtemperature thermostats will be provided in the building. An alarm signal only will be provided by the first thermostat at the first set point. A combination of the first alarm and the second overtemperature thermostat setting will initiate a transfer to standby control.

Lot of Ventilating Equipment for Control House
The control equipment house will be ventilated by the equipment described as follows:

Three air handlers powered by 460V, 3-phase motors. Two of these will provide adequate air changes to remove the heat from the building. The third unit will come on if for any reason the temperature in the house reaches a preset limit.

Ventilation will be thermostatically controlled.

Item	Qty.	
17	2	Iron-Core Filter Reactor - Fifth Harmonic

Reactor assembly comprised of three single-phase reactors stacked vertically.

Features:
Indoor, dry-type, iron-core, convection-cooled in metal enclosure, with removable access panels.
Base designed for skidding or rolling
Provision for lifting
Ground pad
Ambient: 0°C to +40°C
Altitude: 6900 Feet

PROPRIETARY INFORMATION OF GENERAL ELECTRIC COMPANY

Item	Qty.	
		Rating Each reactor: 300 Amp RMS, 60 Hz 1.0 millihenries inductance 20,000 amperes short-circuit rating
18	2	Single-Phase Static Inverters The following applies to each inverter: Output Capacity: 5 KVA Input Voltage: 105 - 140 VDC from customer's station battery Output Voltage: 115V, 60 Hz. Ambient: 0°C to 40°C. Included Options: Automatic static transfer switch Manual bypass switch Automatic retransfer circuit

PROPRIETARY INFORMATION OF GENERAL ELECTRIC COMPANY

DRAWING REFERENCE LIST

Power Conversion/Control

Elementary	228B3432, Sh. OA thru 4AE
Outline	173C3407, Sh. 1
Terminal Boards	228B3433, Sh. 1,2,3,4
Typical Views	188B1894, Sh. 12,20

Power Factor Capacitors

Outline	184C2919

Standby Resistor and Contactor Panel

Outline	171B6845, Sh. 1
Elementary/Assembly	202B5456
Resistor Connections	202B5248

Transformer and Switch (Rapid Electric)

Outline	C10-500-19, Sh. 1
Elementary	C19-0603, Sh. 1

DC Reactor (Rapid Electric)

Outline	C18507-148, Sh. 1
Elementary	B20760-W4, Sh. 1

System Interconnection

Interconnection	173C3415, Sh. 1,2,3,4,5

Tuning Reactor (Rapid Electric)

Outline	C18507-174, Sh. 1
Elementary	B19-0657, Sh. 1

Motor Reference Drawings

Outline, Tag 1A	34C112568
Tag 1B	34C107366
Protective Component Assembly	34C845204
Shaft Sketch	34B475320
Connection Diagram	178C2248

PROPRIETARY INFORMATION OF GENERAL ELECTRIC COMPANY

Control House Drawings

Dura Bilt Products, Inc. Job No. 81-113

Foundation Plan. . . . .Dated 4/9/81
Elevation. . . . . .Rev. 5/12/81

Equipment Layout. . . . . .173C3406, Sh. 1

Motor Control Center

Outline - summary . . . . . .229B7937, Sh. 1, 2.
Elementary. . . . . .228B3432, Sh. 4AA - 4AD.

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TAB 9 HOUSE AIR HANDLING SYSTEM

Index

Bill of Material & List of Publications

House Air Handling System - Description

Control Scheme

Circuitry

PROPRIETARY INFORMATION OF GENERAL ELECTRIC COMPANY

BILL OF MATERIAL & LIST OF PUBLICATIONS

QUANTITY	CATALOG NO.	PUBLICATION	DESCRIPTION
3	Model 25B	CLCH-1M-7	Cabinet Fan including filter base, inlet vanes, and ceiling spring isolators.
3	M944D1017	G0-2292-1	Modulating Motor
3	T92D1048	60-2243-3	Modulating Thermostat
3	Q605D1069	60-2003-1	Damper Linkage
1	T636A1005	71-92213	"High Limit" Thermostat
3	3830A1005	60-2022-3	Filter Flag/Switch
18	FLR496	-	20X25 Filters
27	FLR498	-	20X20 Filters

House Air Handling System - Description

The ventilation system for the house has been designed with the following features and characteristics.

There are three identical Air Handlers. Each Air Handler has a 15 horsepower 1750 RPM, 460 volt, 3 phase motor driving a 500 RPM impeller through a belt drive which will allow a plus or minus 10% on-site adjustment in the final set speed. Air Handler has vane control to allow CFM to be adjusted in the range of 3745 minimum to 22,000 maximum.

Air handler has a filter box attached containing FARR 30/30 medium efficiency filters, quantity six 20X25 and quantity nine 20X20. Filters are of disposable type with recommended change intervals of 2 to 3 months depending on environment. Air Handlers weigh 1800 pounds each and are suspended from building roof trusses.

The position of Air Handlers is given on Revision 10 of the house plan 173C3406. Vendor is the Trane Company.

The controls for each Air Handler are dedicated to that unit only and no cross interlocking is used.

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Control Scheme

Air Handler unit no. 1 will start whenever drive 1 starts. CFM, being drawn into house by unit 1, will be vane-controlled by a proportional thermostat mounted at the air intake to the thyristor cabinet of drive 1. Similarly for unit 2 and drive 2. Unit 3 will come on if, for any reason, the temperature of the air being drawn into either thyristor cabinets reaches 115°F. This could happen in the event no. 1 or no. 2 was not functioning properly or because of a combination of extreme high outside temperature and maximum power being delivered by both drives. For these reasons, unit 3 is a back up for unit 1 or 2 but on rare occasions may be called on for peaking duty. No. 3 will have two thermostats to turn it on or it can be turned on manually. Anti-cycling circuitry is included in the control for this unit. Its vane is controlled by a proportional thermostat mounted at a central strategic point in the room.

Circuitry

The circuitry for this equipment is shown on sheets 4AA, 4AC and 4AD of elementary 228B3432. The starters for the fan motors are housed in the motor control center.

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GENERAL ELECTRIC COMPANY • DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC